

q-wallet

Quantum-Safe Sui

What quantum computers break, what quantum safe means, and how Sui lets an account switch to a quantum-safe key without changing its address. Then we build it.

day1-quantum-safe-wallet.vercel.app

[KEYS, DOLLARS AND BITCOIN · DAY 1 · SUIHUB LAGOS]

RURU (@MINTING_RURU)

Quantum computers: one kind of math

Not a faster laptop. A machine that wins big on a few problems with hidden structure, and one of them is ours.

01 Qubits and interference

Qubits hold blends of 0 and 1. Algorithms make wrong answers cancel out. Only works on structured problems.

02 Shor's algorithm

Turns a public key into its private key, for RSA and every elliptic curve. This is the threat to signatures.

03 Grover's algorithm

Speeds up brute-force search by a square root only. A 256-bit hash keeps about 128 bits of security.

Google, 2026: under 1,200 logical qubits break a 256-bit curve. No machine today can run it.

What breaks and what does not

Keys break. Hashes hold. So your address can stay; the key behind it has to change.

Cryptography	Where you meet it	Against a quantum computer
Ed25519, secp256k1, secp256r1	Signing transactions: wallets, passkeys	Broken: Shor recovers the private key
Groth16 proofs, short-lived keys	zkLogin (Google, Apple sign-in)	Broken: also elliptic-curve math
BLAKE2b-256 hash	Turning a public key into your address	Holds: Grover leaves about 128 bits
AES-256, ChaCha20	Encrypting backups and connections	Holds: Grover halves, still strong
ML-DSA-65	Post-quantum Sui accounts	Holds: no known quantum shortcut

What quantum safe means, and when to act

New math with no known quantum shortcut, standardised by NIST. And a rule of thumb for timing.

Quantum safe means

- Math with no known quantum shortcut
- Lattices (ML-DSA) or hashes (SLH-DSA)
- NIST FIPS 204 and 205, August 2024
- Falcon (FIPS 206) is still not final

When to act: Mosca's rule

- X: how long keys must stay safe
- Y: how long the migration takes
- Z: years until a quantum computer
- If X + Y > Z, you are already late

Every key type Sui accepts

The flag byte in front of every Sui signature says which scheme signed. Flags from @mysten/sui 2.39.0.

Scheme	Flag	Who uses it	Quantum safe?
Ed25519	0	Sui CLI, most Sui wallets	No
secp256k1	1	Bitcoin and Ethereum style keys	No
secp256r1	2	Secure enclaves, HSMs	No
Multisig	3	Teams, treasuries	Only as safe as its keys
zkLogin	5	Sign in with Google or Apple	No
Passkey	6	Face ID, fingerprint, phones	No (secp256r1)
ML-DSA-65	7	Post-quantum accounts	Yes (devnet today)

ML-DSA-65 against Ed25519

Same backup, bigger signatures. Sizes measured on devnet with @mysten/sui 2.39.0.

	Ed25519 (today)	ML-DSA-65 (post-quantum)
Standard	RFC 8032	NIST FIPS 204, level 3
Hard problem	Elliptic curve discrete log	Module lattices
Sui signature flag	0x00	0x07
What you back up	32-byte seed from your phrase	32-byte seed from the same phrase
Public key	32 bytes	1,952 bytes
Signature	64 bytes	3,309 bytes
In the transaction	97 bytes	5,262 bytes
Signing	Deterministic	Hedged: fresh randomness each time

[PART 3 · BEEN HERE BEFORE]

How migrations have gone before

Changing cryptography is slow everywhere. The winners built a way to swap it in advance.

01

The web

SHA-1 weaknesses known in 2005. Browsers dropped it in 2017, weeks before the first public collision.

02

Bitcoin

Keys hidden behind a hash until spent. BIP-360, a draft for post-quantum outputs, came in 2026.

03

Ethereum

Accounts cannot change their key. Plan B (2024): a recovery hard fork after theft begins.

04

The lesson

Build a way to swap the algorithm before you need it, without breaking any address.

How Sui solves it

From Mysten Labs, "Sui's post-quantum signature schemes", 18 August 2026, and our own runs.

01 ML-DSA-65 key

A native key type, flag 7, NIST level 3. Validators check it like any other signature.

02 Same phrase

Derived at m/94'/784' beside today's m/44'/784'. Backups do not change.

03 Address aliases

An address keeps a list of keys allowed to sign. Add the new one, retire the old.

04 Opt-in

Nothing moves until the owner signs. Each account chooses when to switch.

05 SLH-DSA vaults

Hash-based signatures checked in Move, for high-value assets. Mainnet targeted for 2026.

06 A killswitch

A zero-knowledge proof of the old EdDSA seed, kept in reserve. Mysten paper, FC 2026.

Live status: devnet only, for now

Feature flag mldsa65_auth, read from each network's GraphQL protocolConfigs on 9 Nov 2026.

DEVNET — protocol 141

On

mldsa65_auth = true. Q-Wallet runs here.

TESTNET — protocol 140

Off

Planned for the end of 2026.

MAINNET — protocol 141

Off

Q1 2027 at the latest, per Mysten. Opt-in.

TYPESCRIPT

```
// src/lib/sui.ts, readPqFlags(): one request per network
body: JSON.stringify({ query: '{ protocolConfigs { protocolVersion featureFlags { key value } } }' })
const f = c.featureFlags.find((x) ⇒ x.key ═══ 'mldsa65_auth');
// a failed read returns mldsa: null (unknown), never false (off)
```

Path A: a quantum-safe key

From the phrase you already have. Q is a new account, quantum safe from day one; nothing to migrate.

TYPESCRIPT

```typescript
import { Ed25519Keypair } from '@mysten/sui/keypairs/ed25519';
import { MLDSA65Keypair } from '@mysten/sui/keypairs/mldsa65';
import { generateMnemonic, validateMnemonic } from '@scure/bip39';

export function newPhrase() { return generateMnemonic(wordlist, 128); }   // 12 words

export function keysFrom(saved: Saved): Keys {
  if (!validateMnemonic(mnemonic, wordlist)) throw new Error('That is not a valid 12 or 24 word phrase');
  return {
    classic: Ed25519Keypair.deriveKeypair(mnemonic),   // m/44'/784'/0'/0'/0'   address X
    quantum: MLDSA65Keypair.deriveKeypair(mnemonic),   // m/94'/784'/0'/0'/0'   address Q
  };
}
```

Path B: upgrade a wallet, keep its address

X is the address you keep, Q the quantum-safe key. Only the key allowed to sign for X changes.

01

Enable

Turn on aliases, signed by the classic key. X now lists [X].

02

Add Q

add(Q), signed by the classic key, in its own transaction. [X, Q].

03

Send as X

Sender X, signature from Q. Accepted because Q is listed.

04

Retire X

remove(X), signed by Q. [Q]. Only after step 3 worked.

05

Refused

The classic key is now rejected: Required Signature from Q is absent.

[PROOF ON DEVNET]

Proof: the run on the live site

pnpm e2e against the Vercel app, 9 Nov 2026. X = 0x4118…b842, Q = 0xb058…b65c.

	Sender	Signed by	Digest
Fund Q from X	X	Ed25519	`81TjtXSntT5FvttX8DK7X783AqhsjkmnZgN7Y3wGXoWT`
Send with ML-DSA-65	Q	ML-DSA-65	`BRf1jB4yQUgqLmF3TH3ghTL4sGQYGF6KJLpWzf4WkSWc`
Turn on aliases	X	Ed25519	`4T8KmoHZHAptKMRRhxu13JxJycf2yzq2qTQi89jjhfMB`
Add Q as an alias	X	Ed25519	`85khdRWZnRbWH2zrJyChAWa2vJb3kK3L2Mie1pMjxmas`
Send as X	X	ML-DSA-65	`7TZYEQ1yCQjpU4wwkSH1ur1UUdF1zHaPLSEyTc9qDuvx`
Retire the classic key	X	ML-DSA-65	`4h5PHmyxf3EGbrGB5A3iAx4GigzjAb4VPvCoYAvAynWc`
Classic key afterwards	X	Ed25519	`Refused: Required Signature from 0xb058…b65c is absent`

[BUILD IT]

Run Q-Wallet, and what will bite you

Node 22+, pnpm and a browser. No environment variables. The only server code is a devnet faucet relay.

SHELL

```
git clone https://github.com/Nuel-osas/day1-quantum-safe-wallet
cd day1-quantum-safe-wallet
pnpm install
pnpm dev      # http://localhost:5292
pnpm build    # typecheck, then production build
pnpm e2e      # clicks the whole flow on devnet

# the same run against the live site
APP_URL=https://day1-quantum-safe-wallet.vercel.app \
  pnpm e2e
```

01 **Devnet only**
mldsa65_auth is off on testnet and mainnet

02 **5 KB per signature**
5,262 bytes against 97; count it in batches

03 **Random signatures**
Never use a signature as an id

04 **Order matters**
Retire the old key only after Q signed as X

05 **Faucet limits**
Devnet faucet limits per IP; one per pair

06 **Keys in localStorage**
Fine for a devnet class, never real money

[TAKE HOME]

Change the key, keep the address.

Switch while the old key is still safe.

Quantum computers threaten the keys every chain uses, not Sui alone. Sui already has a post-quantum key from the same phrase and aliases to move an account onto it. Devnet today, testnet by the end of 2026, mainnet by Q1 2027. Each account opts in, so prepare now and migrate early.